MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : PCC-CS501 Compiler Design

UPID : 005506

Time Allotted : 3 Hours *Full Marks :70*

The Figures in the margin indicate full marks.

Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer *any ten* of the following : [1 x 10 = 10]

(I) What is postfix SDT?

(II) What is equivalence of type expression?

(III) The actual parameters are not used by the calling procedure. (true/false)

(IV) "goto L" is an unconditional jump(to L) three address instruction.(true/false)

(V) A basic block is a sequence of consecutive statements with single entry/single exit. (true/false)

(VI) Flex is a lexical analyzer generator tool. (True/False)

(VII) Write the name of translator that translates assembly code to relocatable machine code.

(VIII) 12_Name is a lexeme of pattern identifier in C language. (True/False)

(IX) Write the rule for converting left recursive grammar to right recursive grammar.

(X) What is Annotated -parse tree?

(XI) Reduce/Reduce conflict happens during bottom up parsing. (True/False)

(XII) How lexical analyzer recognize a token?

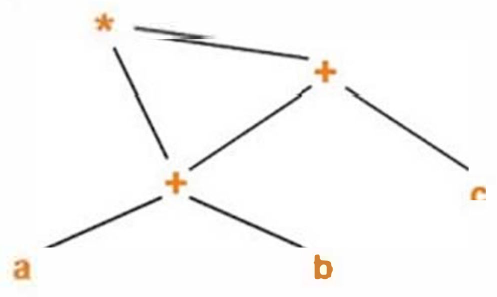
Group-B (Short Answer Type Question)

Answer *any three* of the following : [5 x 3 = 15]

2. Write the difference between synthesized and inherited attributes with examples. [5]

3. Describe the purpose of two pointers of Buffer Pairs in Lexical Analyzer. [5]

4. Convert the DAG into Three address code. [5]



Directed Acyclic Graph

5. Consider the postfix SDT: Here expr, $expr_1$ both are same, and for differentiate between left expr and right expr, we use $expr_1$ in right. [5]

expr → $expr_1$ + term { print(' + ') }

expr → $expr_1$ - term { print(' - ') }

expr → term

term → 0 { print(' 0 ') }

term → 1 { print(' 1 ') }

....

term → 9 { print(' 9 ') }

Draw the parse tree with action embedded for the expression **9 + 4 - 3**.

6. Convert the C code into three address instruction. [5]

while(A[i] ≥ v){ i = i + 1; } where array elements are integers of 4 bytes in sized.

Group-C (Long Answer Type Question)

Answer *any three* of the following : [15 x 3 = 45]

7. (a) Suppose ε-closure(q) is a set of states which are reachable from q with zero or more ε-moves, where q is a state in NFA. You are given a Regular Expression R = (b|a)*baa and the set of input symbols is {a,b}U{ ε }. Convert this Regular Expression R to NFA N. [3]

(b) Convert this NFA N to DFA D using the definition of ε-closure(q). [7]

(c) Convert this DFA D to Minimal DFA. [5]

8. (a) Suppose you have given a grammar of certain kind of statements and first & follow sets: [5]
A→B A' ; A'→ + B A' | ε ; B→ C B' ; B'→ * C B' | ε ; C→ (A) | Id

Non-terminals	First sets	Follow sets
A	(, id	), #
A'	+ , ε	), #
B	(, id	+,), #
B'	* , ε	+,), #
C	(, id	+, *,), #

Where # is an end marker representing the end of input string.
Build the predicting parsing table for the above grammar.

(b) Consider a new set for error recovery of predicting parsing, called synchronizing set of non-terminal A, Is a set where each symbol of synchronizing set of non-terminal A Is taken from follow(A) set. I.e synchronizing(C) = follow(C) = {+, *,), #}. [5]
Instead of writing "error" in M[A, a] in parsing table, you use "syn" in that cell if a belongs to follow(A).
Example: If M[A,)] = "error" in table M, then you use "syn" in M[A,)] cell of M, since ")" belongs to follow(A).
The solution of (b) can be obtained from the following rules given below:
i) If the parser looks up entry M[A, a] and finds that is "error", then the input symbol a is skipped.
ii) If the entry is "sync", then it skip symbols from input until a terminal symbol is seen which belongs to first(A) to continue parsing, if A is the top of the STACK.
III) If a token on top of the STACK does not match the Input symbol, then you pop the token and resume parsing.
Build an error correcting non-recursive version of predicting parsing table for the above grammar.

(c) From the solution of (b), show the behavior of your parser on the following Input: [5]

Sl no	STACK	INPUT	Behavior/Action
1.	**#A**	)id * + id#	....
2.	....	....	....

9. Consider the grammar with productions: https://www.makaut.com [15]
S → A a | b A c | d c | b d a; A → d
Prove that the above grammar Is CLR(1) but not SLR(1) by building the parsing table.

10. (a) Suppose you have been given the three address code of Quick Sort Algorithm [10]

Three address code for Quick Sort					
1	i:= m–1	11	t5:= a[t4]	21	a[t10]:= x
2	j:= n	12	if t5 > v goto (9)	22	goto (5)
3	t1:= 4*n	13	if i >= j goto (23)	23	t11:= 4* i
4	v:= a[t1]	14	t6:= 4*i	24	x:= a[t11]
5	i:= i+1	15	x:= a[t6]	25	t12:= 4*i
6	t2:= 4*i	16	t7:= 4*i	26	t13:= 4*n
7	t3:= a[t2]	17	t8:= 4*j	27	t14:= a[t13]
8	if t3 < v goto (5)	18	t9:= a[t8]	28	a[t12]:= t14
9	j:= j-1	19	a[t7]:= t9	29	t15:= 4*n
10	t4:= 4*j	20	t10:= 4*j	30	a[t15]:= x

Find the leader codes and basic blocks including three address code of the above code.

(b) Build the flow graph for above code. [5]

11. (a) Describe in brief about the cousin of Compiler. [5]

(b) Describe the operation of different phases of Compiler with suitable example. [10]